Module-III

Proposition Logic

A proposition or statement can be defined as a declarative sentence i.e. either True 'T' or False 'F' but not both.

Example:

$3+3=6$ & $3+3=7$, both are the statements but the first one is true and second one is false.

$(x+y)>1$ is not a statement because for some values of the variables, it is true and for others it is false.

Truth Value:

The truth or falsity of a statement is called its truth value. Since only two possible truth values are admitted T or F, this logic is sometimes called two-valued logic.

Note: Questions, exclamations and commands are not propositions.

Types of Propositions:

1. **Atomic**: It is an individual synthetic element consisting of only a single propositional variable or constant. It cannot be sub-divided any further.

2. **Compound**: It is the combination of two or more propositions by means of logical operators (connectives) or by negating a single proposition.

Connectives in the Propositional Logic

Symbol Used	Connective Word	Nature of the Compound statement formed.	Symbol Form	Negation
$\sim$ or $\neg$	not	Negation	$\neg p$	$\sim(\sim p) \equiv p$
$\wedge$	and	Conjunction	$p \wedge q$	$\sim(p) \vee \sim(q)$
$\vee$	or	Disjunction	$p \vee q$	$(\sim p) \wedge (\sim q)$
$\rightarrow$	if... then...	Conditional	$p \rightarrow q$	$p \wedge (\sim q)$
$\leftrightarrow$	if & only if	Biconditional	$p \leftrightarrow q$	$[p \wedge (\sim q)] \vee [q \wedge (\sim p)]$

1. Negation ($\sim$ or $\neg$)

If p is any proposition, the negation of p is denoted by $\sim p$ (or $\neg p$) and read as "not p". It is a proposition that is false when p is true and true when p is false.

p: Ram is healthy.
$\sim p$: Ram is not healthy.

p	~p
T	F
F	T

2. Conjunction (^)

If p & q are two statements, the conjunction of p & q is denoted by $p \wedge q$ and read as "p and q". It is the compound statement.

p: It is raining.
q: It is cold.

$p \wedge q$: It is raining and cold.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

3. Disjunction (v)

If p & q are two statements, the disjunction of p & q is denoted by $p \vee q$ and read as "p or q".

p: Ram is healthy.
q: Ram has blue eyes.

$p \vee q$: Ram is healthy or has blue eyes.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

4. Conditional ($\rightarrow$)

If p & q are two statements, the conditional proposition is denoted as $p \rightarrow q$ and read as "if p then q". It is also called implication.

$p \rightarrow q$

p: Antecedent (Hypothesis)

q: Consequent (Conclusion)

p: Tomorrow is Sunday.
q: Today is Saturday.

$p \rightarrow q$: If tomorrow is Sunday then today is Saturday.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

5. Biconditional ($\leftrightarrow$ or $\Leftrightarrow$ or $\rightleftharpoons$)

If p & q are two statements, the biconditional proposition is denoted as $p \leftrightarrow q$ (or $p \Leftrightarrow q$ or $p \rightleftharpoons q$) and read as "p if and only if q".

p: He swims.
q: The water is warm.

$p \leftrightarrow q$: He swims if and only if the water is warm.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Q. Apply proposition logic for:

(i) $p \wedge (\sim q \vee q)$ (ii) $[\sim(p \vee q)] \vee [\sim p \wedge \sim q)]$

(i) Sol^n:

p	q	$\sim q$	$\sim q \vee q$	$p \wedge (\sim q \vee q)$
T	T	F	T	T
T	F	T	T	T
F	T	F	T	F
F	F	T	T	F

$p \wedge (\sim q \vee q) = T, T, F, F.$

(ii) Sol^n:

p	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$	$\sim(p \vee q) \vee (\sim p \wedge \sim q)$
T	T	T	F	F	F	F	F
T	F	T	F	F	T	F	F
F	T	T	F	T	F	F	F
F	F	F	T	T	T	T	T

$\sim(p \vee q) \vee (\sim p \wedge \sim q) = F, F, F, T.$

Q. Prove the following: $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

Soln:

Statements			LHS		RHS		
p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Hence, $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$ proved

Q. Prove the given: $p \rightarrow q = \sim q \rightarrow \sim p$

Soln:

		LHS	RHS		
p	q	$p \rightarrow q$	$\sim q$	$\sim p$	$\sim q \rightarrow \sim p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Hence, $p \rightarrow q = \sim q \rightarrow \sim p$ proved.

9. Solve the given $[(p \vee q) \vee (\sim p \vee r)] \leftrightarrow (q \vee r)$.

Soln:

p	q	r	$p \vee q$	$\sim p$	$\sim p \vee r$	$(p \vee q) \vee (\sim p \vee r)$	$q \vee r$
T	T	T	T	F	T	T	T
T	T	F	T	F	F	T	T
T	F	T	T	F	T	T	T
T	F	F	T	F	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	F

$[(p \vee q) \vee (\sim p \vee r)] \leftrightarrow (q \vee r)$
T
T
T
F
T
T
T
F

Q. Solve the following equation:

Ram is unhealthy and he has not blue eyes.

Sol^n:

The given is,

$$(\sim p) \wedge (\sim q)$$

p	q	$\sim p$	$\sim q$	$(\sim p) \wedge (\sim q)$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

$(\sim p) \wedge (\sim q)$ = F, F, F, T.

Q. Solve: $(p \wedge \sim q \wedge \sim r) \vee (q \wedge r)$

Sol^n:

p	q	r	$\sim q$	$\sim r$	$p \wedge \sim q \wedge \sim r$	$q \wedge r$	$(p \wedge \sim q \wedge \sim r) \vee (q \wedge r)$
T	T	T	F	F	F	T	T
T	T	F	F	T	F	F	F
T	F	T	T	F	F	F	F
T	F	F	T	T	T	F	T
F	T	T	F	F	F	T	T
F	T	F	F	T	F	F	F
F	F	T	T	F	F	F	F
F	F	F	T	T	F	F	F

Tautology

A compound proposition that is always true for all possible truth value of its variable.

In other words, if the final column contain only T then it is called Tautology.

Example:

$$\sim p \vee p$$

p	$\sim p$	$\sim p \vee p$
T	F	T
F	T	T

Contradiction

A compound proposition that is always false for all possible value of its variable.

In other words, if the last column contain only F then it is called Contradiction.

Example:

$$\sim p \wedge p$$

p	$\sim p$	$\sim p \wedge p$
T	F	F
F	T	F

Contingency

A proposition function which is sometime True and sometime False, then it is known as contingency

Example:

$p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Satisfiability

A compound proposition that contain at least one True value in the last column is called as Satisfiability.

Example:

$p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

9. Show that $(p \vee q) \vee (\sim p \vee r) \rightarrow (q \vee r)$ is Contingency and Satisfiable.

Soln:

p	q	r	$\sim p$	$p \vee q$	$\sim p \vee r$	$(p \vee q) \vee (\sim p \vee r)$	$q \vee r$
T	T	T	F	T	T	T	T
T	T	F	F	T	F	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	F	T	T	T
F	F	F	T	F	T	T	F

$(p \vee q) \vee (\sim p \vee r) \rightarrow (q \vee r)$
T
T
T
F
T
T
T
F

Hence, the given is contingency and satisfiable.

Q. ~~Show that~~ Determine $(\sim p \leftrightarrow q) \leftrightarrow (q \leftrightarrow r)$ is Tautology and Satisfiable.

Soln:

p	q	r	$\sim p$	$\sim p \leftrightarrow q$	$q \leftrightarrow r$	$(\sim p \leftrightarrow q) \leftrightarrow (q \leftrightarrow r)$
T	T	T	F	F	T	F
T	T	F	F	F	F	T
T	F	T	F	T	F	F
T	F	F	F	T	T	T
F	T	T	T	T	T	T
F	T	F	T	T	F	F
F	F	T	T	F	F	T
F	F	F	T	F	T	F

Hence, it is Satisfiable but not Tautology.

Some Other Conditionals

If p & q are two proposition function then some other conditional proposition related to $p \to q$ are:

1. **Converse**

Converse of $p \to q$ is $q \to p$.

2. **Contrapositive**

Contrapositive of $p \to q$ is $\sim q \to \sim p$.

3. **Inverse**

Inverse of $p \to q$ is $\sim p \to \sim q$.

Q. Convert the following statement into converse and contrapositive.

$$(p \vee q) \to (p \wedge q)$$

Soln:

Converse is $(p \wedge q) \to (p \vee q)$.

Contrapositive is $\sim(p \wedge q) \to \sim(p \vee q)$.

Converse Contrative

p	q	p∨q (let A)	p∧q (let B)	A → B	B → A	~B → ~A
T	T	T	T	T	T	T
T	F	T	F	F	T	F
F	T	T	F	F	T	F
F	F	F	F	T	T	T

More Topics Remaining